MODULE 4

MONTE CARLO SAMPLING

Monte Carlo Sampling on Noisy 2D Blobs

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1. OBJECTIVE

The main objective of this assignment is to estimate a probability using Monte Carlo sampling on a synthetic noisy two-dimensional blobs dataset.

The objectives are:

- To generate two synthetic clusters with different centre positions.
- To define a vertical decision boundary at X = 0.
- To estimate the probability that a point from Cluster A crosses the boundary.
- To compare estimates obtained using different sample sizes and random seeds.
- To study how changing the noise level affects cluster overlap and boundary-crossing probability.
- To compare Monte Carlo estimates with the corresponding theoretical probability.

2. TOOLS AND DATASET DESCRIPTION

I. Tools and technologies

- Google Colab
- Python
- NumPy – generation of random samples and numerical calculations
- Matplotlib – visualization of clusters and experimental results
- SciPy – calculation of theoretical normal-distribution probabilities

II. Dataset description

A separate external dataset is not required for this experiment. The dataset is generated programmatically using NumPy's random normal distribution. It consists of two synthetic clusters, each containing 1,000 points in the initial visualization.

Property	**Description**
Dataset name	Noisy 2D Blobs
Dataset type	Synthetic, generated using NumPy
Number of clusters	2 (Cluster A and Cluster B)
Initial samples	1,000 points per cluster; 2,000 total

Number of features	2 (X and Y coordinates)
Cluster A centre	(-1, 0)
Cluster B centre	(1, 0)
Initial noise / standard deviation	0.8
Random seed	42
Decision boundary	Vertical line X = 0

III. Dataset visualization

Cluster A is generated around (-1, 0), while Cluster B is generated around (1, 0). The vertical dashed line at X = 0 is used as the decision boundary. The experiment counts Cluster A points whose X coordinate is greater than zero.

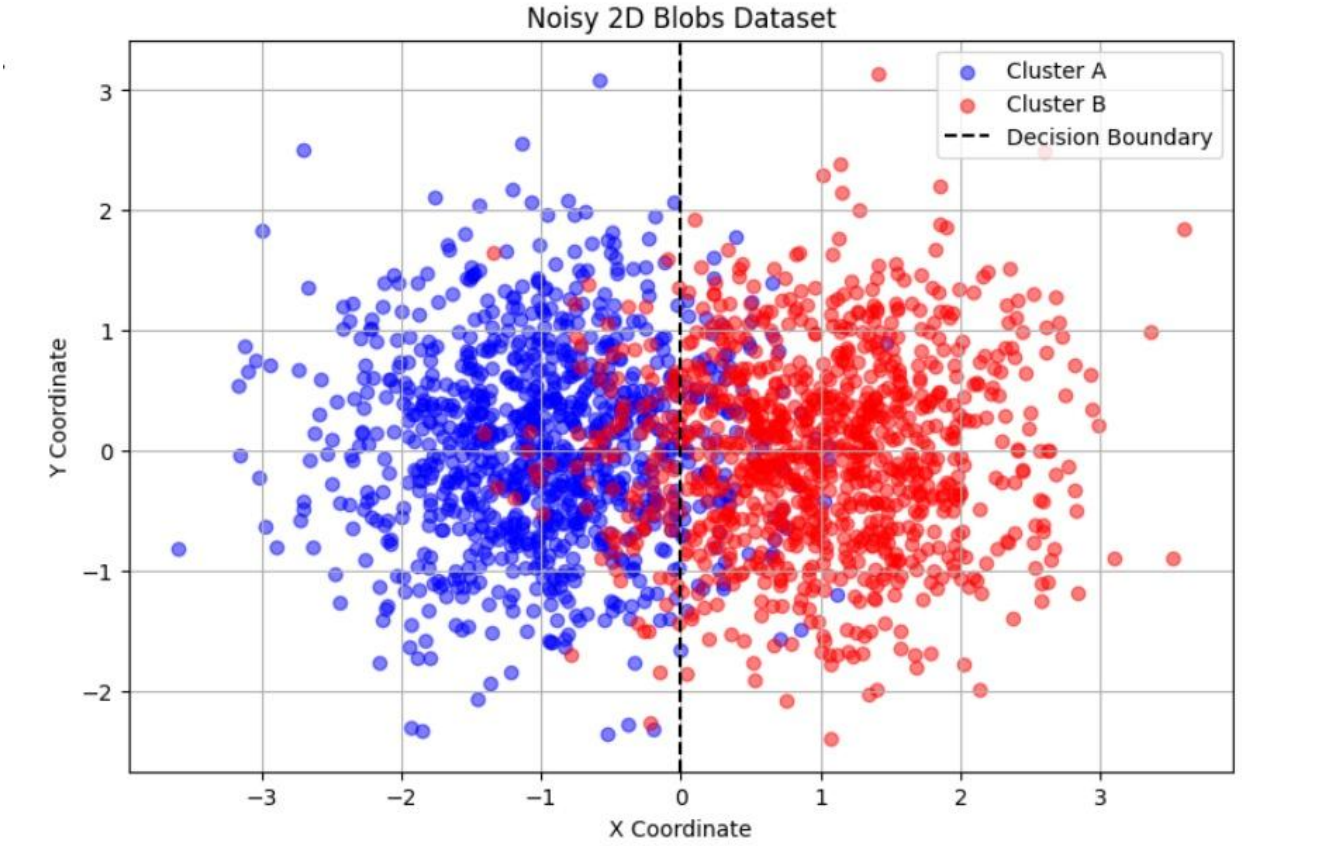


Figure 1. Synthetic noisy 2D blobs with the decision boundary X = 0.

4. MONTE CARLO ALGORITHM AND MATHEMATICAL BASIS

Monte Carlo sampling estimates a probability by generating repeated random samples and calculating the fraction of samples that satisfy a specified condition.

In this experiment, the event is that a point from Cluster A crosses the boundary X = 0. The estimated probability is:

$$\hat{P}(X > 0) = \text{Number of Cluster A points with } X > 0 \,/\, \text{Total number of Cluster A points}$$

Cluster A has an X coordinate distributed as a normal random variable with mean -1 and standard deviation σ (the noise level). Therefore, the theoretical probability is:

$$\mathbf{P(X > 0) = 1 - \Phi(1 / \sigma)}$$

Here, Φ represents the cumulative distribution function of the standard normal distribution. For σ = 0.8, the theoretical probability is approximately 0.10565.

The Monte Carlo estimate is a boundary-crossing probability for Cluster A. It is not a trained-classifier accuracy or a measured misclassification rate.

5. IMPLEMENTATION

I. Generating the synthetic data

The generate_blobs() function sets the random seed and generates two arrays using NumPy's normal distribution. Each array contains two-dimensional points.

II. Monte Carlo estimation

The monte_carlo_estimation() function generates the clusters, counts the points in Cluster A for which X > 0, and divides this count by the requested sample size.

III. Theoretical comparison

SciPy's normal cumulative distribution function is used to calculate the theoretical probability. The absolute error is the absolute difference between the estimated and theoretical probabilities.

IV. Experiments

Three experiments are conducted: varying the sample size, varying the random seed, and varying the noise level. Graphs are generated in the notebook to visualize the results.

Main program code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm
# Generate two noisy 2D clusters

def generate_blobs(n_samples=1000, noise=0.8, seed=42):
    np.random.seed(seed)
    A = np.random.normal(
        loc=[-1, 0],
        scale=noise,
        size=(n_samples, 2)
    )

    B = np.random.normal(
        loc=[1, 0],
        scale=noise,
        size=(n_samples, 2)
```

```
    )
    return A, B
# Monte Carlo probability estimation
def monte_carlo_estimation(n_samples, noise=0.8, seed=42):

    A, B = generate_blobs(n_samples, noise, seed)

    crossing_points = np.sum(A[:, 0] > 0)

    probability = crossing_points / n_samples

    return probability


# Execute the Monte Carlo experiment
n_samples = 10000
noise = 0.8
seed = 42

estimated_probability = monte_carlo_estimation(
    n_samples, noise, seed
)

# Calculate theoretical probability
theoretical_probability = 1 - norm.cdf(1 / noise)

# Calculate absolute error
absolute_error = abs(
    estimated_probability - theoretical_probability
)

# Display results
print("Monte Carlo Estimate:", estimated_probability)
print("Theoretical Probability:", theoretical_probability)
print("Absolute Error:", absolute_error)
```

6. RESULTS AND PERFORMANCE EVALUATION

I. Monte Carlo and theoretical probability

Measure	**Value**
Sample size	10,000
Noise level	0.8
Monte Carlo estimate	0.10810
Theoretical probability	0.10565
Absolute error	0.00245

Error as percentage points of probability	0.245%

II. Effect of sample size

Sample size	Estimated probability
100	0.06000
1000	0.10200
10000	0.10810
50000	0.10812

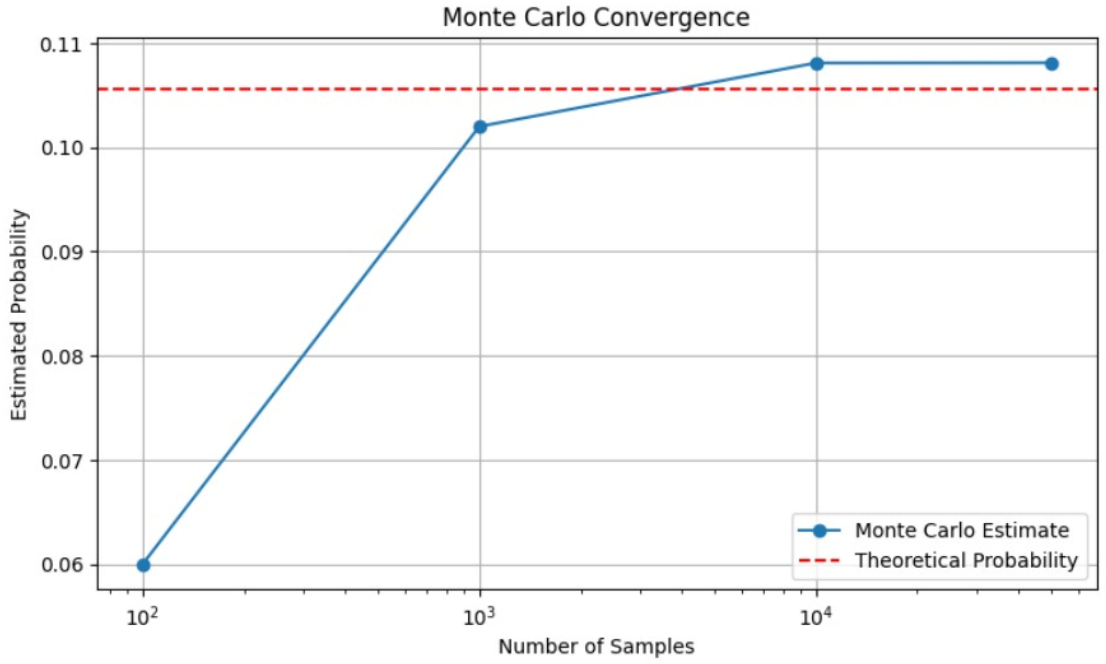


Figure 2. Monte Carlo estimate compared with theoretical probability for different sample sizes.

The estimate becomes more stable as the sample size increases. The estimates for 10,000 and 50,000 samples are close to the theoretical probability, while the estimate from 100 samples shows larger random variation.

III. Effect of random seeds

Random seed	Estimated probability
10	0.10330
20	0.10620
30	0.10490
40	0.10230
50	0.10540

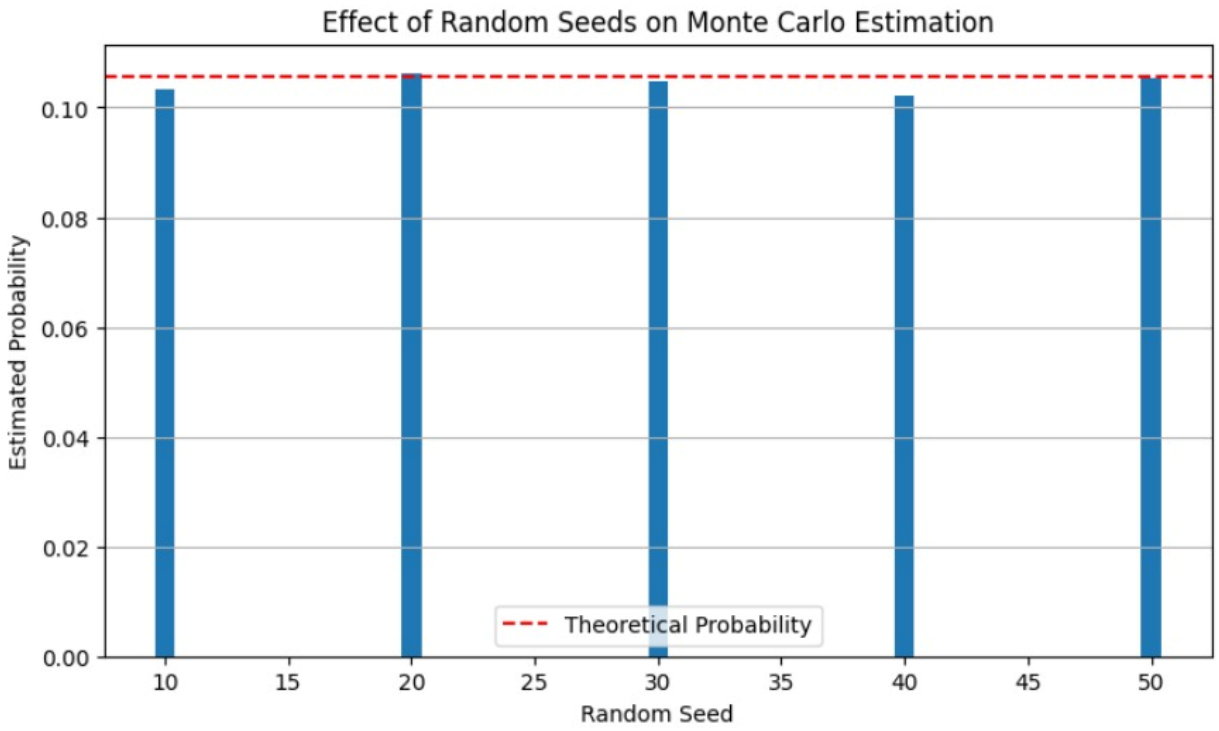


Figure 3. Estimated probability for different random seeds (10,000 samples each).

The estimates vary slightly with the random seed because each seed produces a different sequence of random samples. The values remain in a relatively narrow range around the theoretical probability.

IV. Effect of noise level

Noise level	Monte Carlo estimate	Theoretical probability
0.3	0.00020	0.00043
0.5	0.02400	0.02275
0.8	0.10810	0.10565
1.0	0.16370	0.15866
1.5	0.25720	0.25249

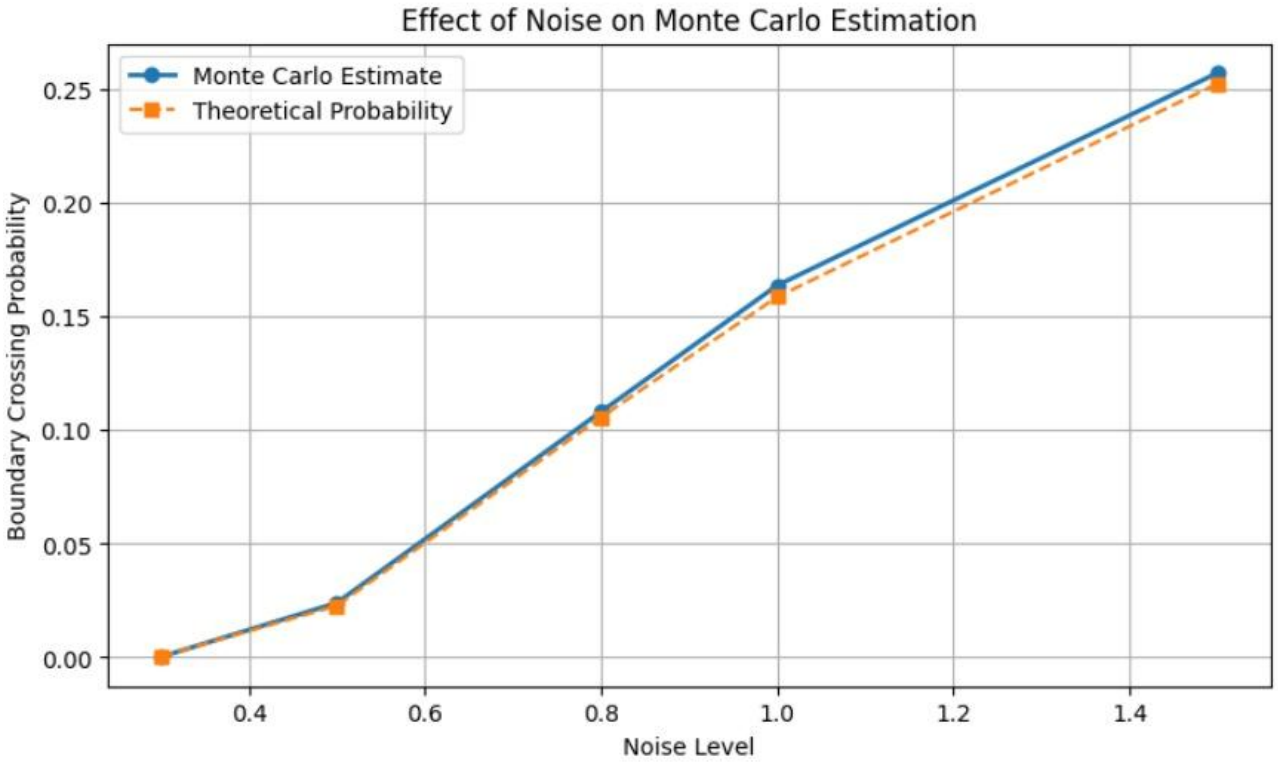


Figure 4. Effect of noise on boundary-crossing probability.

As the noise level increases, the clusters become more spread out. More points from Cluster A cross the boundary X = 0, so the estimated boundary-crossing probability increases.

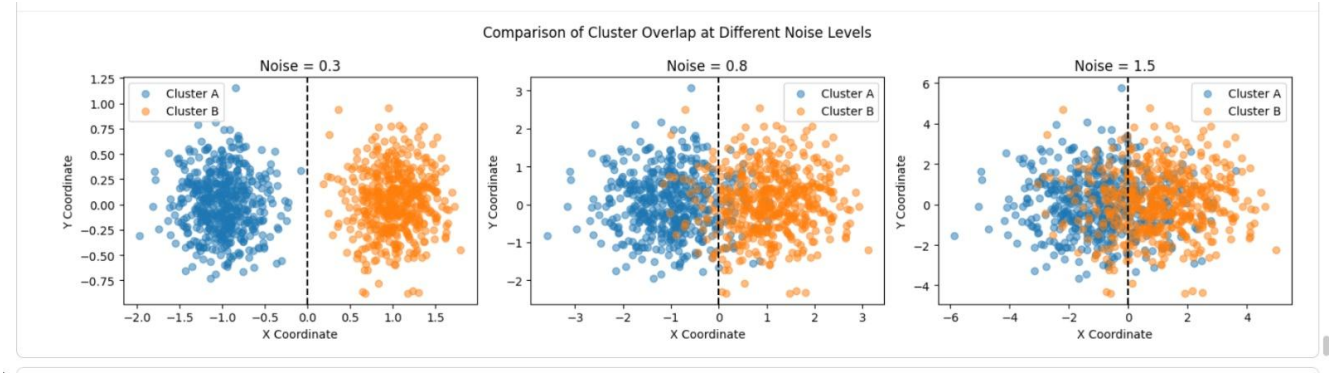


Figure 5. Visual comparison of cluster overlap at noise levels 0.3, 0.8 and 1.5.

V. Error and standard error analysis

Samples	Estimate	Theory	Absolute error	Standard error
100	0.06000	0.10565	0.04565	0.03074
1000	0.10200	0.10565	0.00365	0.00972
10000	0.10810	0.10565	0.00245	0.00307
50000	0.10812	0.10565	0.00247	0.00137

The standard error decreases as the sample size increases. This indicates that estimates based on larger samples generally have lower sampling variability. The absolute error need not decrease perfectly at every step because Monte Carlo estimates are random.

7. DISCUSSION

The experiment demonstrates how Monte Carlo sampling can estimate a probability from generated data. With noise 0.8 and 10,000 samples, the estimated probability was 0.10810, compared with a theoretical value of approximately 0.10565. The absolute difference was 0.00245.

Increasing the sample size generally improves the stability of the estimate. Changing the random seed produces small variations, which are expected in a random sampling experiment. Increasing the noise spreads the clusters and increases the probability that a point from Cluster A crosses the boundary.

The theoretical probability provides a reference for checking the Monte Carlo estimate. Small differences between the two values are expected because the Monte Carlo result is based on a finite random sample.

8. CONCLUSION

This assignment successfully implemented Monte Carlo sampling on a synthetic noisy 2D blobs dataset. Two clusters were generated using NumPy, and the probability of Cluster A points crossing the boundary $X = 0$ was estimated.

The experiment compared different sample sizes, random seeds and noise levels. The results showed that larger sample sizes provide more stable estimates, random seeds cause small variations, and higher noise increases cluster overlap and boundary-crossing probability. The Monte Carlo estimate for 10,000 samples at noise 0.8 was 0.10810, close to the theoretical probability of 0.10565.

9. LIMITATIONS

- The experiment uses a synthetic dataset rather than real-world observations.
- Monte Carlo estimates can vary because of finite random sampling.
- The analysis uses a fixed vertical decision boundary at $X = 0$.
- Only the boundary-crossing probability for Cluster A is estimated; no classifier is trained.

10. REFERENCES

1. NumPy documentation, Random sampling: https://numpy.org/doc/stable/reference/random/

2. SciPy documentation, scipy.stats.norm:
https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.norm.html

3. Matplotlib documentation: https://matplotlib.org/stable/